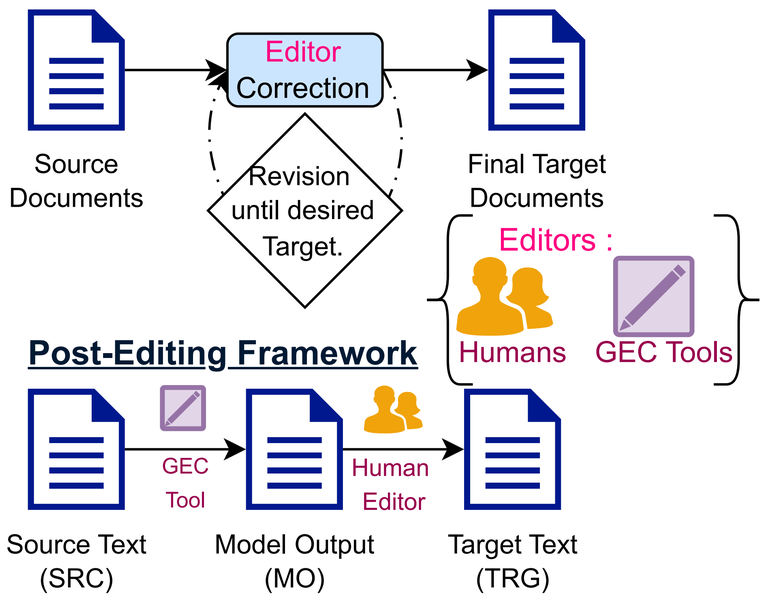
Time Is Effort: Estimating Human Post-Editing Time for Grammar Error Correction Tool Evaluation

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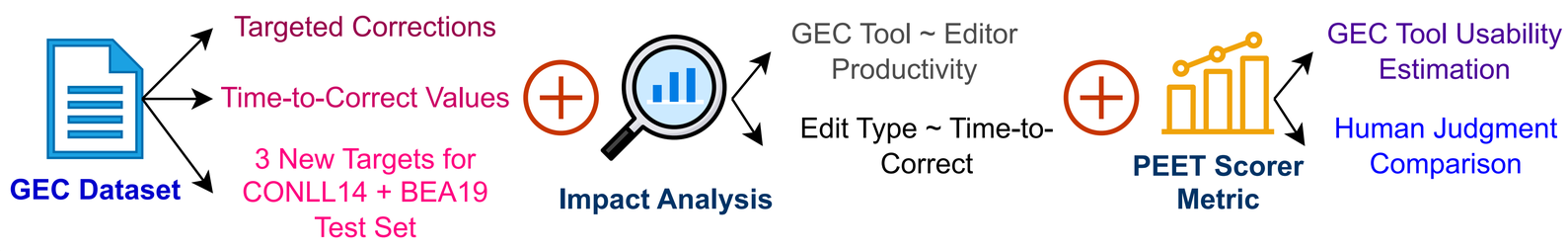
1. Concept



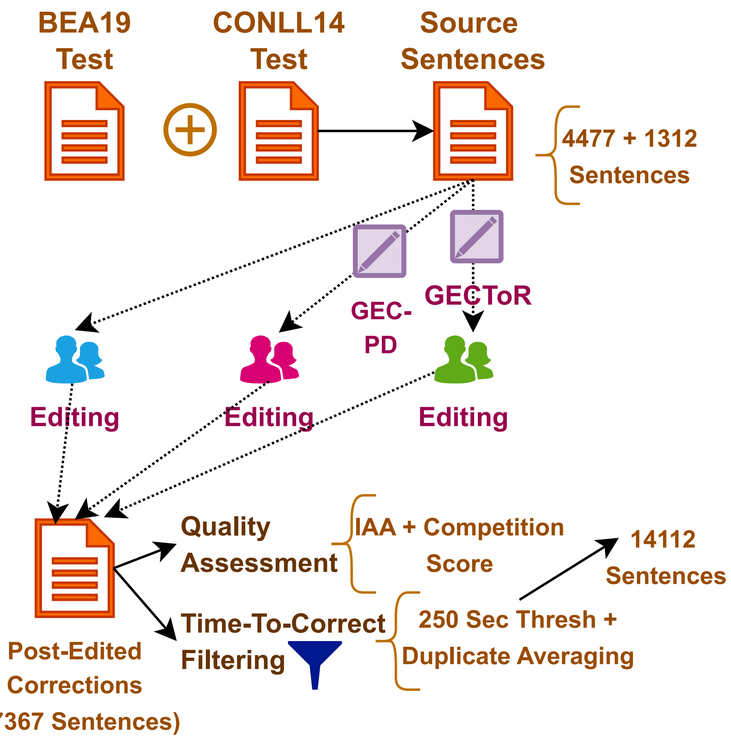
2. Summary

- GEC Tool **Post-Editing (PE)** follows the Text Editing framework. This allows for generating **targeted corrections** and **human-centric GEC Tool evaluation**.
- **Temporal Effort** (time-to-correct) can capture the **total PE effort** (technical + cognitive effort).
- Consider the **edit type** and **closest Target** correction for GEC-Tool **usability estimation**.

3. This Work



4. Data Collection Framework



5. Dataset Quality Assessment

Candidate Set	BEA19 Test (Precision : Recall : $F_{0.5}$)	CONLL14 Test (Precision : Recall : $F_{0.5}$)
Source Sentence	-	-
Source Sentence EC	45.30 : 66.08 : 48.34	49.05 : 60.45 : 50.97
GECToR Output	66.81 : 58.42 : 64.94	63.97 : 45.94 : 59.31
GECToR Output EC	48.24 : 71.38 : 51.59	50.50 : 61.09 : 52.31
GEC-PD Output	66.20 : 61.48 : 65.20	64.06 : 44.92 : 59.03
GEC-PD Output EC	47.33 : 70.54 : 50.66	52.17 : 60.86 : 53.71
GRECO Model Output	86.45 : 63.13 : 80.50	79.36 : 48.69 : 70.48

Table 1: Quality Scores of the 2 GEC Tools output prediction, target Editor Corrections (EC) and the State-of-the-Art GEC Tool - GRECO on the official BEA19 and CONLL14 competition.

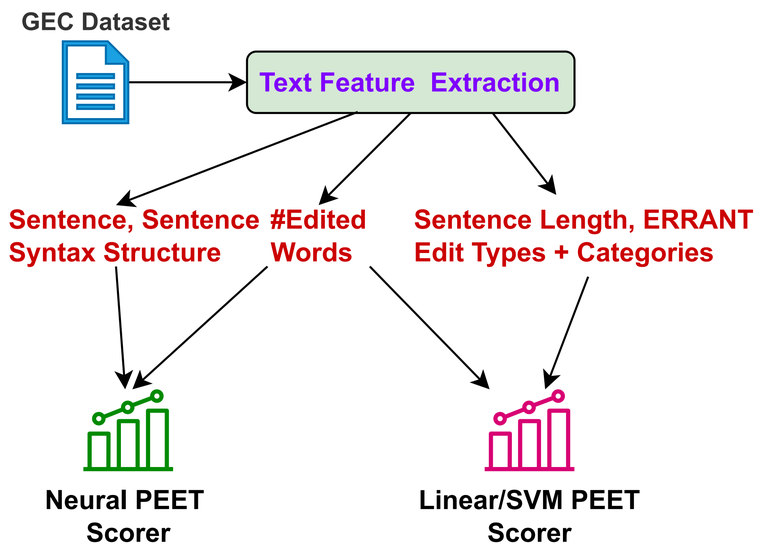
- We see significantly higher Recall Scores for the EC (more corrections made), like other public references.
- Inter-Annotator Agreement (IAA) Scores for our EC are better than other public references (Table 4 in paper).

- **Impact of GEC Tools on Text Editing :**

Editing Time Average ~ 4 seconds saved per sentence when using GEC Tools for first-pass Text Editing.

EC Target Improvement GEC Tool output EC has better scores than the Source Sentence EC.

6. PEET Scorer Methodology



7. PEET Scorer Insights

- Linear/SVM Regression Models perform as well as Neural Models.
- Analyzing standardized Linear Regression PEET Scorer coefficient weights makes feature impact interpretable.
- MO-TRG feature selection is sufficient and on par with SRC-MO-TRG features.
- Considering the ERRANT edit type hierarchies gives - small (4), medium (25), and big (55) features to train PEET regression models.
- We consider the best Linear Regression model (medium) features for edit impact analysis.

8. PEET Scorer Model Performance

Stat. Model	Edit Feat. Level	r	MAE
Linear Reg.	4	0.559	18.92
	25	**0.565**	18.74
	55	0.563	18.75
SVR Linear	4	0.558	16.40
	25	0.564	16.19
	55	**0.565**	16.15

Table 2: Average PEET Scorer performance for the Stat. Models over 50 runs (different train-test data seed). Results - Pearson Correlation (r) and Mean Absolute Error (MAE) loss.

9. Impact of Edits on Temporal Effort

Model Features	Regression Coefficient	Model Features	Regression Coefficient
OTHER	10.15	**ADV**	1.79
PUNCT	4.55	**VERB:FORM**	1.66
PREP	4.03	**WO**	1.63
VERB	3.37	**VERB:SVA**	1.16
Sentence Correct	-3.31	**PRON**	1.10
NOUN	3.23	**ADJ**	0.97
DET	3.08	**CONTR**	0.78
NOUN:NUM	2.52	**VERB:INFL**	0.63
VERB:TENSE	2.35	**PART**	0.47
ORTH	2.34	**ADJ:FORM**	0.39
CONJ	2.03	**NOUN:INFL**	-0.30
MORPH	1.89	**NOUN:POSS**	0.25
SPELL	1.87	-	-

Table 3: The standardized regression coefficients of the LR model trained on the ERRANT - medium (25) edit features to measure the impact of each feature on PEET estimation.

10. PEET Scorer $\sim$ Human Judgment

Human Judgment Ranking	PEET Metric		WER	
	ρ	r	ρ	r
Grundkiewicz - C14 (EW)	0.48	0.26	0.28	0.18
SEEDA - C14 - All (TS)	0.18	0.63	0.18	0.65
SEEDA - C14 - NO (TS)	-0.1	-0.27	-0.1	-0.33
Napoles - FCE	**-0.96**	-0.94	**-0.96**	-0.88
Napoles - Wiki	**-0.71**	-0.63	**-0.93**	-0.88

Table 4: The correlation of our PEET model with 3 human rankings (HJR) and the Word Edit Rate (WER) metric. A high negative correlation indicates a lower time-to-correct and WER score. Corresponding to a higher human judgment ranking.

- We observe a strong correlation with HJR that correlates with the WER - Technical Effort.

11. Limitations and Future Work

- PE annotation for evaluating GEC tool performance can be expensive and time-consuming.
- We utilize automated edit-type features from the ERRANT toolkit. Descriptive edit features (e.g., from Grammar Error Explanation research) can be used as possible features for the PEET Scorer.
- Perceived PE effort does not always correlate with actual effort. Studying GEC cognitive PE effort and its relation to technical and temporal PE effort could explain this better.
- Exploring GEC PE studies for other languages can also be interesting.